

1. What is a file? Explain the need of file in Python.

A **file** is a collection of data stored permanently on a computer. In Python, files are used to **store and retrieve data** such as text, numbers, records, etc.

Need of files in Python:

- To store data permanently.
- To read previously stored data.
- To save program output for future use.
- To handle large amounts of data.
- To share data between programs.

Example:

```
file = open("data.txt", "w")  
file.write("Hello Python")  
file.close()
```

2. State and explain the different modes of opening a file in Python.

Python provides different modes to open a file:

Mode Meaning

r	Read the file
w	Write to the file
a	Append data to the file
x	Create a new file
r+	Read and write
w+	Write and read
a+	Append and read
b	Open file in binary mode

Explanation:

r – Read mode

Used to read data from a file. The file must already exist.

w – Write mode

Used to write data to a file. If the file already exists, its old content is removed.

a – Append mode

Used to add new data at the end of an existing file. Old data is not removed.

x – Create mode

Used to create a new file. It gives an error if the file already exists.

b – Binary mode

Used for binary files such as images, videos, and PDF files.

3. State the difference between Write and append mode in opening a file in python.**Write (w)**

Writes data to a file.

Existing content is deleted.

New data starts from the beginning.

Creates the file if it does not exist.

Append (a)

Adds data to a file.

Existing content is preserved.

New data is added at the end.

Also creates the file if it does not exist.

Example:**Write mode:**

```
file = open("data.txt", "w")  
file.write("Hello")  
file.close()
```

After using w:

Hello

Append mode:

```
file = open("data.txt", "a")
```

```
file.write(" Python")
```

```
file.close()
```

If the file contains:

Hello

After using a:

Hello Python